Skills Summary

One-Step Mass and Liquid Volume Problems

Use this reference while completing your worksheet or when studying.

The units. Mass is how heavy something is: grams (g) and kilograms (kg). Liquid volume is how much liquid fills a container: millilitres (mL) and litres (L).

Choosing the one step. “In all”, “altogether”, “put together” → **add**. “Left”, “poured out”, “taken away”, “how much more” → **subtract**. Equal groups → **multiply**. Shared equally → **divide**.

The unit rule. The answer to a measurement question always carries a unit, and it is the same unit the givens were measured in.

1. Mass problems (grams and kilograms)

Write both masses down with their unit before you choose an operation. Masses combine exactly like plain numbers — what the unit does is tell you the answer is still a mass.

Example: A tin of beans has a mass of 320 g. A tin of corn has a mass of 180 g. What is the mass of both tins together?
Step 1. “Together” means the two masses are put into one amount, so this is one addition.
Step 2. $320 + 180$: make a friendly number first, $320 + 80 = 400$, then add the 100 that is left. The two tins have a mass of **500** g.

Try it: A melon has a mass of 460 g and a pear has a mass of 140 g. What is their total mass, in grams?

check: **600** g

2. Liquid volume (millilitres and litres)

Liquid poured *out* of a container makes the amount smaller, so the answer must be *less* than what you started with. That size check catches most wrong answers before the grader does.

Example: A bottle holds 900 mL of water. Sam pours 250 mL into a glass. How much water is left in the bottle?
Step 1. Water leaves the bottle, so the one step is a subtraction, and the answer has to come out smaller than 900 mL.
Step 2. $900 - 250$: take away 200 to get 700, then take away 50. The bottle still holds **650** mL.

Try it: A jug holds 750 mL of lemonade. Ada pours out 400 mL. How much lemonade is left, in millilitres?

check: **350** mL

3. Read the measurement, then take one step

When the measurements sit in a table, the first job is *reading*: find the two rows the question names and ignore the rest. Then take the single step the question asks for.

Example: A record table gives bag A a mass of 700 g and bag B a mass of 450 g. How much heavier is bag A than bag B?
Step 1. "How much heavier" compares two masses, so read those two rows and subtract the smaller from the larger.
Step 2. $700 - 450 = 250$. Bag A is **250** g heavier than bag B.

Try it: The same table gives jug A 600 mL of water and jug B 375 mL. How much more water is in jug A, in millilitres?

check: **225** mL

(!) Watch out: an answer bigger than what you started with is a warning sign on a "poured out" or "taken away" problem — it usually means the two numbers were added. And never drop the unit: 450 on its own does not say whether the water is 450 mL or the melon is 450 g.